Milestone - 5 Report

Introduction:
This milestone includes using ensemble models. This experiment was done in two parts using 5 fold validation, with out of fold model training, then later ensembling them with equal weights.

1. 5-fold 2 stage training DaBERTa (8(2e-5) + 8(5e-6)) - fold 4 discarded due to no learning
2. 5-fold 1 stage training DaBERTa (8(5e-6))

Performance:

Model	Validation Loss	Prediction performance (MAP@3)
Baseline (sample_submission.csv)	-	0.3525
TF-IDF	0.29617	0.32294
Word2vec	0.34792	0.30423
sBERT	0.42308	0.38653
BART-MNLI	0.56583	0.49985
DaBERTa (3 epochs)	-	0.62177
RoBERTa (3 epochs)	-	0.49958
DaBERTa (8 epochs at 2e-5)	-	0.74064
DaBERTa (16 epochs at 2e-5)	-	0.71945
DaBERTa (8(2e-5) + 8(5e-6))	-	0.73316
DaBERTa (8(2e-5) + 5(5e-6))	-	0.74397
RAG model (MiniLM-v6)	-	0.68536
DaBERTa (8(2e-5) + 5(5e-6)) 5-fold (without the discarded fold)	-	0.75436
DaBERTa (8(5e-6)) 5-fold	-	0.73815

Conclusion:
The experiments ran with two different sets of epochs, while the two stage model showed a higher average validation loss between all folds compared to the single stage fine tuning which showed a much lower and consistent lower validation loss, the two stage tuning model, performed higher on the kaggle leaderboard. While the single stage tuning didn’t show great

signs of learning during the first stage, it converged well during the second stage. With the single stage model, it converged almost immediately, but performed worse, this may be a sign that the 2-stage model settled on a different minima than the single stage one, which was more suited for the testing dataset. My explanation is that the first stage higher learning rate probably helped the model wander the loss landscape slightly more freely before settling on a minima that better represents the learnings from the training set.